AP PRECALCULUS · TOPIC 1.6

Polynomial Functions and End Behavior

Section 1 — Multiple Choice
Circle the letter of the correct answer **SOLVED**

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $f(x) = -2x^5 + 3x^2 - 7$. Which statement describes the end behavior of f?

A. $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$

B. $f(x) \to -\infty$ as $x \to \infty$ and $f(x) \to \infty$ as $x \to -\infty$

C. $f(x) \to -\infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$

D. $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to \infty$ as $x \to -\infty$

B. Leading term $-2x^5$: odd degree, negative coefficient, so the right end falls and the left end rises.

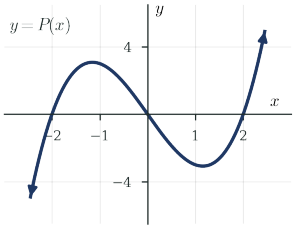
2 Let $g(x) = -x^2(2x-1)(x+4)$. What is $\lim_{x \to -\infty} g(x)$?

A. ∞ **B.** 0

C. -2 **D.** $-\infty$

D. Leading term $-x^2 \cdot 2x \cdot x = -2x^4$: even degree, negative coefficient, so both ends fall.

3 The graph of a polynomial P is shown. Which could be the degree and leading coefficient of P?



A. Degree 3, positive leading coefficient

B. Degree 3, negative leading coefficient

C. Degree 2, positive leading coefficient

D. Degree 4, negative leading coefficient

A. Left end down and right end up means odd degree with a positive leading coefficient; two turning points means degree at least 3.

4 A polynomial function has degree 6. What is the greatest number of turning points its graph can have?

A. 6 **B.** 7

C. 5 **D.** 3

C. A polynomial of degree n has at most $n-1$ turning points.

5 Let $h(x) = 5x - x^3 + 2x^4 - 9$. Which statement is true?

A. $h(x) \to -\infty$ as $x \to -\infty$ and $h(x) \to \infty$ as $x \to \infty$

B. $h(x) \to \infty$ as $x \to -\infty$ and as $x \to \infty$

C. $h(x) \to \infty$ as $x \to -\infty$ and $h(x) \to -\infty$ as $x \to \infty$

D. $h(x) \to -\infty$ as $x \to -\infty$ and as $x \to \infty$

B. Reorder: $2x^4 - x^3 + 5x - 9$. The leading term $2x^4$ has even degree and a positive coefficient, so both ends rise.

6 A polynomial q has three turning points, and $q(x) \to -\infty$ as $x \to \infty$ and as $x \to -\infty$. Which could be the degree and leading coefficient of q?

A. Degree 4, negative leading coefficient

B. Degree 4, positive leading coefficient

C. Degree 3, negative leading coefficient

D. Degree 5, negative leading coefficient

A. Both ends fall, so the degree is even and the leading coefficient is negative; three turning points needs degree at least 4.

7 Which function satisfies $f(x) \to \infty$ as $x \to -\infty$ and $f(x) \to -\infty$ as $x \to \infty$?

A. $f(x) = x^3 + 6x$ **B.** $f(x) = -x^4 + 6x$

C. $f(x) = x^4 - 6x$ **D.** $f(x) = -x^3 + 6x$

D. Left end up and right end down is odd degree with a negative leading coefficient.

8 Let $f(x) = (x-2)(x+1)(3-x)$. Which statement is true?

A. $f(x) \to -\infty$ as $x \to -\infty$ and $f(x) \to \infty$ as $x \to \infty$

B. $f(x) \to \infty$ as $x \to -\infty$ and as $x \to \infty$

C. $f(x) \to \infty$ as $x \to -\infty$ and $f(x) \to -\infty$ as $x \to \infty$

D. $f(x) \to -\infty$ as $x \to -\infty$ and as $x \to \infty$

C. The leading term is $x \cdot x \cdot (-x) = -x^3$: odd degree, negative coefficient.

AP PRECALCULUS · TOPIC 1.6 · SECTION 2

Polynomial Functions and End Behavior

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

At the outdoor music festival, the number of visitors inside the grounds, in hundreds, is modeled by $N(t) = -0.5t^3 + 8t^2 + 2t$, where t is the number of hours after the gates open and $0 \le t \le 16$.

(a) Write $\lim_{t\to\infty} N(t)$. Explain why the model cannot describe the grounds for all $t \ge 0$. DOK3 2 pt

$\lim_{t\to\infty} N(t) = -\infty$ (leading term $-0.5t^3$). A negative number of visitors is impossible, so the model fails for large t.

(b) Find the greatest number of visitors predicted on $0 \le t \le 16$ and the time it happens. Give t and $N(t)$ to three decimal places. DOK3 2 pt

Maximum at $t \approx 10.790$ hours, $N \approx 324.865$ hundreds, about $32{,}486$ visitors.

(c) Find the first time after opening when the model predicts $N(t) = 0$ again. Give your answer to three decimal places. DOK3 2 pt

Solve $-0.5t^2 + 8t + 2 = 0$: $t \approx 16.246$ hours (the positive root).

Question 10 No calculator · 6 points

Let $r(x) = -x^3(x-2)^2(x+1)$.

(a) State the degree of r and the sign of its leading coefficient. DOK3 2 pt

Leading term $-x^3 \cdot x^2 \cdot x = -x^6$: degree 6, leading coefficient -1 (negative).

(b) Write $\lim_{x\to-\infty} r(x)$ and $\lim_{x\to\infty} r(x)$. DOK3 2 pt

Even degree, negative: $\lim_{x\to-\infty} r(x) = -\infty$ and $\lim_{x\to\infty} r(x) = -\infty$.

(c) Explain why the graph of r has at least one turning point and at most five. DOK3 2 pt

Both ends fall, so the graph must turn at least once; a degree 6 polynomial has at most $6 - 1 = 5$ turning points.

Answer key (Section 1): 1B 2D 3A 4C 5B 6A 7D 8C